

Jayacharan R

 Pollachi , TN
  jayacharan.2023@vitstudent.ac.in
  096550 33521
  jayacharan-r
  JayacharanR

Education

Vellore Institute Of Technology, BTech in Computer Science (Data Science) – Vellore July 2023 – present
 • CGPA : 9.30 / 10.00

Achyuta Public School, CBSE, Highschool in Bio-Mat – Dindigul, TN, India June 2017 – Apr 2023
 • Grade: AISSCE: 89.6% , AISSE: 98%

Projects

Intelligent Media Lifecycle Manager (Python/Rust) Jan 2026 – present

Architected a high-performance CLI tool to organize and deduplicate massive media archives using Python and Rust.

- Engineered a CLI tool with Rust-accelerated Perceptual Hashing (pHash) to detect visual duplicates robust to edits and resizing.
- Implemented a journaled transaction system that logs file operations to JSON, enabling atomic execution and complete state rollback/undo.

WikiMind: Production-Grade Agentic Hybrid RAG & Knowledge Engine June 2026 – present

- Architected the pipeline using LangGraph to orchestrate two-stage local retrieval (dense and sparse) over a Qdrant vector database, integrating NeMo Guardrails for LLM middleware and Langfuse for observability and telemetry.
- Engineered advanced retrieval modalities including entity-based Knowledge Graph traversal (spaCy/NetworkX), temporal versioning for time-travel queries, and automated evaluation harnesses measuring Recall@K/MRR, while extending offline inference to mobile environments utilizing ONNX-quantized embeddings and SQLite-vss.

Experience

AI-ML Intern, Tata Consultancy Services - TCS – Chennai, TN, India June 2026 – July 2026

- Architected a secure RAG pipeline using FastAPI and LangChain, implementing custom Aegis guardrails to enforce contextual relevance and mitigate prompt injection risks.
- Established full-stack LLM observability with real-time dashboards, instrumenting execution traces to monitor retrieval latency, token utilization, and generation quality.

SAP Analytics Cloud Intern, Blueprint Technologies Pvt Ltd – Bengaluru, KA, India May 2025 – July 2025

- Developed responsive dashboards in SAP Analytics Cloud by conducting exploratory data analysis, cleaning data, and integrating live Google Drive feeds with time-based metrics to enable real-time performance monitoring.
- Implemented predictive models (linear regression, classification, time series) and advanced dimension-level calculations to deepen analytical insights, while successfully resolving geo-mapping issues by debugging latitude-longitude ID conflicts.

Awards

Winner - Roboverse Ideathon 2025, IEEE Robotics and Automation Society, VIT, Vellore Jan 2025

- Engineered an AI-driven surveillance prototype using YOLO (Ultralytics) for real-time anomaly detection and human movement tracking in CCTV footage.
- Demonstrated scalable crime prevention capabilities by implementing automated monitoring, behavioral analysis, and security alerts with comprehensive event logging.

Skills

Programming: Python, Java, C, C++ , Bash, YAML, HTML, CSS, JavaScript

AI/ML Frameworks: PyTorch, LangChain, Langfuse, RAG, LLM Observability, LLM Guardrails, Quntization, Fine-tuning, Hugging-Face Embeddings, ONNX Runtime, Qdrant (Vector DB), Redis, NetworkX (Knowledge Graphs), Hybrid Search, Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, NVIDIA NeMo Guardrails

Operating System: Arch Linux, Windows

Software: SAP Analytics Cloud, Excel, Sheets, Git Version Control, AWS Architecture, Docker

Language: English (Professional proficiency), Tamil (Native proficiency), Japanese & Hindi (Limited proficiency)

Courses & Certifications

AWS Certified Solutions Architect – Associate	Dec 2025 – Dec 2028
Microsoft Certified: Azure AI Engineer Associate - AI 102	Nov 2025 – Nov 2026
Microsoft Certified: Azure Data Fundamentals - DP 900	Oct 2025